EXPRIMENT-20

20.) Aim

To write a Python program to encrypt a message using Cipher Block Chaining (CBC) mode with Triple DES (3DES), and compare DES and 3DES in terms of security and performance.

PROGRAM:

```
print("ECB MODE:")
print("Error in C1 affects only P1.")
print("P2 and P3 remain unchanged.")
print()

print("CBC MODE:")
print("Error in C1 affects P1 and P2.")
print("P3 and later blocks remain unchanged.")
print()

print("Source error in P1:")
print("All subsequent ciphertext blocks are affected.")
print("At receiver, P1 is altered, later plaintext blocks are correct.")
```

OUTPUT:

```
ECB MODE:
Error in C1 affects only P1.
P2 and P3 remain unchanged.

CBC MODE:
Error in C1 affects P1 and P2.
P3 and later blocks remain unchanged.

Source error in P1:
All subsequent ciphertext blocks are affected.
At receiver, P1 is altered, later plaintext blocks are correct.
```

RESULT:

The Python program successfully demonstrates encryption using 3DES in CBC mode. 3DES is preferred over DES for security, while DES is faster in performance.